



Amrita School of AI
COIMBATORE CAMPUS

RELEASED WEEK 6 · DUE WEEK 9

Assignment 2: Make One Kernel Scale

Take a serial kernel to a well-scaling parallel one, isolating every change, and explain the efficiency you did not get.

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RELEASED

Week 6

DUE

End of week 9

WEIGHT

1.5% of the course (assignments total 6%)

MARKS

30

WORK

Individual

OFFERING

Odd Semester 2026-27

The task

Choose **one** kernel from the list below, or propose your own by email before week 7. Take it from a correct serial implementation to a well-scaling OpenMP one, and account for every factor of performance along the way.

KERNEL		THE INTERESTING DIFFICULTY
A	Sparse matrix - vector product (CSR)	Irregular access, load imbalance across rows
B	Histogram of a large image, 256 bins	Contention and false sharing
C	k - means assignment and update step	A reduction over cluster centroids
D	Barnes - Hut style force accumulation, direct $O(N^2)$ form	Triangular loop, load imbalance
E	2 - D Jacobi stencil	Memory bound, first - touch and NUMA

Part A: Serial first (8 marks)

Before any threads.

A1. Correct serial implementation, verified against an independent reference.

A2. Roofline placement: compute f by hand, use your Lab 3 machine numbers, state the attainable ceiling and what fraction of it you reach.

A3. At least two serial optimisations, each measured on its own, each with a prediction stated beforehand. Loop order, blocking, avoiding redundant work, data-layout changes and compiler flags all count.

A4. State your final serial GFLOP/s and percentage of attainable.

PITFALL · THE ORDER IS PART OF THE MARK

Parallelising a cache-hostile kernel gives you many cores waiting on memory. Submissions that skip Part A and go straight to `#pragma omp parallel for` lose the Part A marks and usually score badly in Part B as well, because their baseline is wrong.

Part B: Parallelise (12 marks)

- B1.** Add OpenMP. State the data-sharing decision for **every** variable, and use `default(none)`.
- B2.** Correctness under concurrency: run your test at 1, 2, 4, 8 and 16 threads, repeatedly, and show the results are identical. A single run is not evidence.
- B3.** Scaling study following the Lab 3 protocol: T_1 from the **best serial** version, thread pinning, warm-up, five runs, minimum and spread. Report S , E and Karp-Flatt e at each point, with both plots.
- B4.** Schedule: try at least three schedules and report efficiency for each. Justify your choice from the shape of the work, not by trying things at random.
- B5.** Identify and fix at least one of: a race, false sharing, a contended critical section, load imbalance, or a NUMA first-touch problem. Show the before-and-after measurement.

Part C: Explain the gap (10 marks)

- C1.** At your largest thread count, what is your efficiency? Where did the rest go? Attribute it, with evidence, among: serial fraction, synchronisation, memory bandwidth saturation, load imbalance, NUMA effects.
- C2.** Is your Karp-Flatt e constant or rising? State which, and what it implies about the cause.
- C3.** Compute the maximum speedup Amdahl allows given your measured e . How far from it are you, and what would you attack next?
- C4.** Would more cores help? Answer with a number and a reason, not an opinion.

What to submit

`assign2-<R0LLN0>.pdf` plus your source. The report must include the roofline placement, the optimisation table with predictions, both scaling plots, the schedule comparison, the before-and-after for Part B5, and a reproducibility statement.

MARKS	FOR
8	Part A: serial correctness, roofline placement, isolated optimisations
12	Part B: correct parallelisation, repeated correctness evidence, a proper scaling study
10	Part C: an evidence-based attribution of the lost efficiency

Policies

Late: 80 percent within 48 hours, then not accepted without documented cause. Collaboration: discuss freely, write your own code and report; all submissions are compared for similarity. AI tools: permitted for learning; you must be able to explain every line and every number, and a viva decides where a submission and your explanation disagree.